NANYANG JUNIOR COLLEGE
Year 2 Preliminary Examination

H2 GEOGRAPHY 9173/01

Paper 1 17 Sept 2025

3 hours

Additional Materials: Insert

READ THESE INSTRUCTIONS FIRST

If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.
Write your Centre number, index number and name on the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO **NOT** WRITE ON ANY BARCODES.

Answer both questions in **Section A**.
Answer two questions in **Section B**, one from each cluster.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Section A

Answer **both** questions.

Cluster 1: Development, Economy and Environment

1 Resource 1 shows a map of the cobalt supply chains. Resource 2 shows the distribution of mineral deposits in the Democratic Republic of Congo (DRC), Africa. Resource 3 shows the area of land cover classes of interest (km^2) and global price of cobalt (USD/T) from 2009 to 2021 in DRC. Resource 4 shows a photograph of a cobalt artisanal small-scale mining (ASM) site in DRC.

(a) With reference to Resource 1, describe the main global flows in the cobalt supply chain. [4]

Award 1 mark for each description of the main global flows in the cobalt supply chain, to a maximum of 4 marks.
Award a maximum of 1 additional mark for further development of each description, where applicable.

Possible responses include:

- Resource 1 shows that the cobalt supply chain begins with extraction in the Democratic Republic of Congo (DRC), where raw cobalt is mined. [1]
- The cobalt is then exported to countries like China, where it is refined and processed into intermediate components. [1]
- These parts are then shipped to battery manufacturers in countries such as South Korea and Japan, where they are assembled into batteries. [1]
- The batteries are then distributed globally including USA and Europe, completing the flow from raw material to end-user. [1]

(b) With reference to Resource 1, explain how the cobalt supply chain demonstrates interdependence between countries and regions through trade flows. [6]

Award 1 mark for each explanation of how the cobalt supply chain demonstrates interdependence between countries and regions through trade flows, to a maximum of 6 marks.
Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Possible responses include:

- The supply chain demonstrates that global manufacturing is critically dependent on a single region for the extraction of raw cobalt. [1] The resource clearly illustrates this by showing the chain originates exclusively in the DRC, which is identified as the location of artisanal mines, with a major trade route for the raw material flowing out from there. [1 additional mark]

- Countries that extract raw cobalt are interdependent with nations that possess the specialisation and comparative advantage to process it into a usable form. [1] This is shown by the trade flow of raw cobalt from the DRC directly to China, which the map identifies as a primary hub for smelting and refining, making each nation essential to the other for this stage of production. [1 additional mark]

- The global production of high-tech batteries and components relies on the specialized manufacturing capabilities concentrated in a few key Asian countries. [1] The map shows that after refining, the cobalt is used by component producers and battery manufacturers located in China, South Korea, and Japan, to produce the sophisticated components required for modern electronics and electric vehicles. [1 additional mark]

- Manufacturing economies are highly dependent on the demand and purchasing power of major consumer regions to buy the finished products. [1] This final link is visualized by the trade arrows flowing from the Asian manufacturing hubs to Europe and the U.S.A., which are depicted on the map as the key end-markets with consumer brands. [1 additional mark]

(c) Describe the distribution of mineral deposits in the Democratic Republic of Congo as shown in Resource 2. [4]

Award 1 mark for each description of the distribution of mineral deposits in the Democratic Republic of Congo, to a maximum of 4 marks.
Award a maximum of 1 additional mark for further development of each description, where applicable.

Possible responses include:

- The DRC's mineral deposits show clear regional clustering. [1]

- Highest clustering along the eastern border regions which include gold, tin ores and niobium tantalum ores. [1]

- Highest Representation – Gold and Tin ores deposits which widely scattered across the eastern border regions, especially near Uganda, Rwanda, and Burundi. [1]

- Lowest Representation – Manganese and Coal shown only at isolated points (e.g., manganese near the southeastern region, coal around the southern border). [1]

- Copper and Cobalt Ores are concentrated heavily in the southern region near the border with Zambia. [1]

- Diamond deposits are concentrated in the central-southern regions, particularly near Kananga and Mbuji-Mayi. [1]

(d) Cite data from Resource 3 to describe the trend in the global price of cobalt (USD/T) between 2010 and 2021 in the Democratic Republic of Congo. [4]

Award 1 mark for each description using data from Resource 3 on the trend in the global price of cobalt (USD/T) between 2010 and 2021 in the Democratic Republic of Congo, to a maximum of 4 marks.
Award a maximum of 1 additional mark for further development of each description, where applicable.

Possible responses include:

- Following an initial price of approximately $40,000/T in 2010, the price of cobalt entered a period of general decline to its lowest point around 2016, dropping to approximately $28,000/T. [1]

- From 2016, the price of cobalt experienced a dramatic and steep increase. It surged to its peak in 2018, reaching approximately $75,000/T. [1] This represents the highest price point for cobalt in the entire 2010-2021 period shown in the graph.

- After reaching its peak in 2018, the price of cobalt fell sharply. By 2020, the price had dropped to approximately $32,000/T, returning to levels seen before the major surge. [1]

- In the final year of the observed period, the price of cobalt showed a significant recovery, rising from its 2020 low to approximately $52,000/T in 2021.

(e) With reference to Resources 3 and 4, explain the challenges that extractive industries pose to the sustainable development of the Democratic Republic of Congo. [6]

Award 1 mark for an explanation of the challenges that extractive industries pose to the sustainable development of the Democratic Republic of Congo, to a maximum of 6 marks.
Award 1 additional mark for further development of the challenges that extractive industries pose to the sustainable development of the Democratic Republic of Congo, where applicable.
Award a maximum of 3 marks for explanation of the challenges that extractive industries pose to the sustainable development of the Democratic Republic of Congo of either economic, social or environmental dimension only.

Possible responses include:

Extractive industries in the Democratic Republic of Congo (DRC) pose severe and interconnected environmental, social, and economic challenges that hinder its sustainable development.

Environmental Challenges

- The mining sector inflicts lasting damage on the DRC's environment through widespread land degradation and deforestation. Extractive activities strip the land of its natural cover, permanently altering the landscape. [1] Resource 3 hints at this impact through its land cover data, showing the presence of "Bare land" and

"Exposed rock" which increase at the expense of "Vegetation" due to mining. [1 additional mark]

- Furthermore, artisanal mining causes severe local environmental degradation and pollutes critical water sources. [1] Resource 4 clearly shows workers submerged in heavily turbid, muddy water, a direct result of soil erosion and the washing of ore which contaminates the local watershed, harming both ecosystems and communities. [1 additional mark]

Social Challenges

- Socially, the extractive industry in the DRC is plagued by extreme labour exploitation, including the widespread use of child labour. This practice robs children of their education and future, trapping families in an intergenerational cycle of poverty. [1] This is starkly illustrated in Resource 4, where several children are visibly working alongside adults in the dangerous and physically demanding conditions, a clear violation of their rights and a significant barrier to the nation's social progress. [1 additional mark]

Economic Challenges

- Economically, the DRC's heavy reliance on cobalt exposes the national economy to the severe shocks of global price volatility. [1] Resource 3 illustrates this instability, showing the cobalt price soaring to a peak around 2018 before plummeting sharply in the following years. [1 additional mark]

- Despite extracting high-value minerals, workers in the artisanal mining sector often remain trapped in a cycle of extreme poverty. [1] The reliance on rudimentary tools like simple buckets and the necessity for children to work, as seen in Resource 4, indicates that the economic returns for their hard labour are insufficient to lift them out of poverty or foster community development. [1 additional mark]

(f) Suggest three ways in which the Democratic Republic of Congo could promote economic diversification to reduce dependence on extractive industries. [6]

Award 1 mark for each suggestion in which the Democratic Republic of Congo could promote economic diversification to reduce dependence on extractive industries, to a maximum of 3 marks.
Award a maximum of 1 additional mark for further development of each suggestion, where applicable.

Possible responses include:

- The DRC should prioritize investment in modernizing and expanding its agricultural sector to achieve food sovereignty and grow exports. [1] By supporting smallholder farmers with better infrastructure, access to credit, and modern techniques, the country can leverage its vast arable land to create rural jobs and generate revenue from cash crops like coffee and cocoa. [1 additional mark]

- The nation must develop domestic industries to process its natural resources before they are exported. [1] Building local refineries for minerals and processing plants for timber and agricultural goods would allow the DRC to capture a much larger share of the global value chain, creating higher-skilled jobs and increasing export earnings. [1 additional mark]

- The DRC should develop its immense hydroelectric potential to power domestic growth and become an energy exporter. [1] Fully realizing projects on the Congo River, such as the Inga Dams, would provide the stable and affordable electricity needed to attract manufacturing investment and sell surplus power to neighbouring countries. [1 additional mark]

- The government could promote the country's unique natural heritage by investing in the sustainable tourism sector. [1] By ensuring security and developing infrastructure for its world-renowned national parks like Virunga, the DRC can create service-based employment and generate income from activities like gorilla trekking, providing a strong economic incentive for conservation. [1 additional mark]

- The DRC can foster a vibrant digital economy by investing in telecommunications infrastructure and technology education. [1] Expanding internet connectivity and supporting tech start-ups can unlock growth in digital services, mobile finance, and software development, creating a modern economic sector independent of resource extraction. [1 additional mark]

Cluster 2: Tropical Environments

2 Resource 5 shows a type of mass movement in the humid tropics. Resource 6 shows a type of mass movement in the arid tropics. Resource 7 shows a system of slope processes. Resource 8 shows a satellite image of a fluvial landform in the Amazon, Brazil.

(a) With reference to Resources 5 and 6, compare the characteristics of the two types of mass movements shown. [4]

Award 1 mark for each comparison of the characteristics of the two types of mass movements, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each comparison, where applicable.

Possible responses include:

Similarities

- Similar mechanism - both are downslope movements of weathered material under the influence of gravity, representing slope failure processes. [1]

Differences

- Nature of movement – rotational slide (Resource 5) vs rock fall (Resource 6). [1] Rotational slide occurs where movement takes place along a curved slip plane resulting in the backward rotation of material while rock fall occurs where the movement is sudden and free-falling with individual rock fragments breaking off and falling vertically off a steep cliff face (Resource 6). [1 additional mark]

- Speed – rock fall is expected to be faster than rotational slide due to the vertical cliff face and the lack of vegetation cover to hold the rocks in place [1]

- Type of weathered material involved – fine clay/kaolin material (Resource 5) vs individual blocky masses (Resource 6) [1]

(b) Using Resource 7, explain the role of climate in contributing to the occurrence of the mass movement shown in Resource 5. [6]

Award 1 mark for each explanation of the role of climate in contributing to the occurrence of the mass movement, to a maximum of 6 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Possible responses include:

- Mass movement occurs when the safety factor is less than 1 (that is, Shear Strength < Shear Stress).

- The humid tropical climate drives intense chemical weathering that fundamentally reduces the inherent shear strength of the slope material by converting solid rock into weak clays and soil. [1] As shown in Resource 7, climate is a direct cause of weathering, a process that has created the deep profile of low shear strength material (clay-rich soil) visible in the landslide scarp in Resource 5, making it unable to resist the forces acting upon it. [1 additional mark]

- Heavy rainfall, a key feature of the humid tropical climate, increases the shear stress on the slope through the added weight of water while simultaneously causing a critical reduction in the soil's shear strength due to rising pore water pressure. Resource 7 links climate to slope hydrology; this interaction ultimately led to the failure seen in Resource 5, which occurred when the shear stress exceeded the drastically lowered shear strength of the saturated ground. [1 additional mark]

- The climate supports lush tropical vegetation which, while helping to prevent surface erosion, adds significant weight (shear stress) to the underlying weak material and is unable to prevent large, deep-seated landslides like in Resource 5. [1] Resource 7 depicts the relationship between climate and vegetation affecting slope stability, and Resource 5 shows how entire trees were carried along with the sliding block, indicating their roots were not deep enough to anchor the soil to the stable bedrock below. [1 additional mark]

(c) Suggest <u>one</u> physical factor and <u>one</u> human factor that may have contributed to the occurrence of the mass movement shown in Resource 5. [4]

Award 1 mark for each suggestion of one physical factor and one human factor that may have contributed to the occurrence of the mass movement, to a maximum of 2 marks. Award a maximum of 1 additional mark for further development of each suggestion, where applicable.
Award a maximum of 2 marks for either one physical or one human factor only.

Possible responses include:

Physical:

- The presence of weak geological materials, which possess low internal shear strength, allows for failure to occur along a distinctive curved, concave shear plane. [1] This type of failure is common in two different geological settings: either in deep, uniform (homogeneous) materials like a thick layer of clay, or in a layered structure where stronger, permeable rock rests on top of weak, impermeable layers like clay or shales. [1 additional mark]

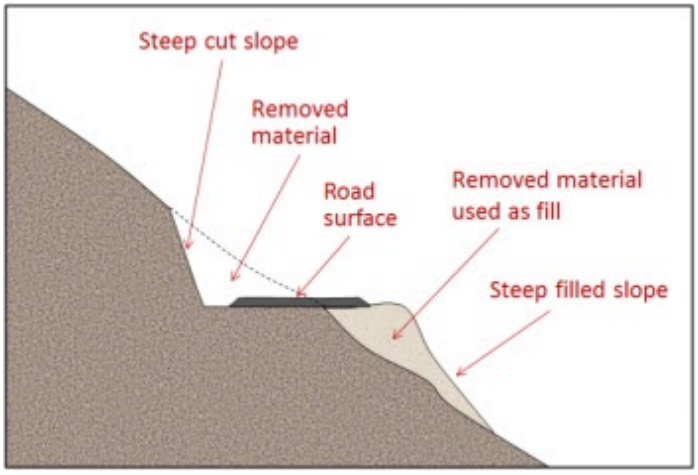
- The steepness of the hill increases the downslope force of gravity acting on the soil and rock, which creates shear stress. [1] Steep slopes are more prone to mass movements because the shear stress is greater, making the slope less stable. [1 additional mark]

- Seismic tremors due to earthquakes could have triggered the slide by suddenly increasing the shear stress on the slope. [1] Ground shaking can also loosen rock formations or cause liquefaction in saturated soil, which rapidly reduces a slope's stability and strength. [1 additional mark]

Human:

- The slope may have been undercut by human excavation, possibly for a quarry or mineral extraction, which would have removed crucial support from its base. [1] Such excavation alters the natural geometry and steepens a slope beyond its angle of

repose, reducing its shear strength and making it much more susceptible to failure. [1 additional mark]



- The clearing of vegetation reduces a slope's stability by removing the root systems that bind soil particles together, which lowers the soil's overall shear strength. Without this natural reinforcement and protective cover, the ground becomes weaker and more susceptible to erosion and saturation from rainfall, increasing the likelihood of a landslide. [1 additional mark]

(d) Describe the main characteristics of the fluvial landform as shown in Resource 8. [4]

Award 1 mark for each description of the characteristics of the fluvial landform, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each description, where applicable.

Possible responses include:

- The landform in Resource 8 is a meandering river with prominent meander bends, like those at X. There are also straight portions of the channel like those at Y. [1]
- There are narrow necks and point bars (or point bar deposits) visible, especially at X, indicating lateral erosion on the outer bends and deposition on the inner bends. [1]
- Oxbow lakes or abandoned meanders can be seen (middle-middle ground), suggesting cut-off processes. [1]
- The channel appears sinuous with floodplains on either side, indicating a low gradient and mature stage of the river. [1]
- There is also dense vegetation on both sides of the river except on the side of the point bar deposit, where land seems to be newly formed. [1]

(e) Using Resource 8 explain how the channel may change over time at sections (X) and (Y). [6]

Award 1 mark for each explanation of how the channel may change over time, to a maximum of 6 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Possible responses include:

At section X:

- The fastest flow (thalweg) is directed against the concave bank at X, causing lateral erosion through hydraulic action and abrasion [1]. Continued erosion leads to the formation of a river cliff and a tighter meander bend. [1 additional mark]
- Over time, the neck of the meander may be breached during floods, forming an oxbow lake and straightening the channel locally.[1]

At section Y:

- Flow here is more uniform and parallel, so the channel tends to be more stable compared to bends.[1]
- However, deposition of fine sediments may still occur along the channel bed in the form of riffles, causing vertical accretion and potential channel shallowing [1].
- As water flows over the riffles, the thalweg tended to be shifted to the sides to allow lateral erosion to take place causing the straight section at Y to meander over time. [1]

Overall, these processes demonstrate the dynamic equilibrium of the river, with continuous erosion, transportation, and deposition shaping the landform.[1]

(f) With reference to Resource 8, explain how human activities may modify the landform. [6]

Award 1 mark for each explanation of how human activities may modify the landform, to a maximum of 6 marks. Award a maximum of 1 additional mark for further development of each description, where applicable.

Possible responses include:

- River engineering works such as channel straightening, embankments, or levees can prevent meander migration and reduce floodplain deposition. [1] This changes the form of the river and is often irreversible. [1 additional mark]

- Deforestation or agriculture in the floodplain increases surface runoff and sediment load, accelerating siltation of channels and increasing deposition within the channel.[1] This can lead to the formation of braided stream when sediments get deposited at the middle of the channel and form channel bars. [1 additional mark]

- Urbanization with impermeable surfaces increases flood risk and may lead to increased discharge that will lead to increased erosion.[1] With increased erosion, there will be more erosional landforms like river cliffs and pools. Braided channels that are not anchored by vegetation might be washed away with high discharge. [1 additional mark]

- Dams upstream can reduce sediment transport and change the erosion-deposition balance, affecting meander development and oxbow lake formation.[1] The failure of dams will also change channel morphology drastically. [1 additional mark]

Section B

Answer **either** question 3 **or** question 4 and answer **either** question 5 **or** question 6.

Cluster 1 Development, Economy and Environment

3 'Human activities have reduced the environment's ability to provide ecosystem services.' Evaluate this statement. [20]

Candidates could approach the question by making a judgement on the extent to which human activities have negatively impacted the ability of the environment to provide ecosystem services, considering provisioning, regulating, cultural and supporting services. Candidates may analyse these impacts through key mechanisms such as habitat destruction, pollution of the natural environment, and the introduction of invasive species, examining how these vary across spatial and temporal scales. Candidates could also consider contexts where human activities have enhanced or restored ecosystem services through conservation, restoration projects or technological innovation.

Candidates could frame their judgement around the relative significance of human activities in degrading ecosystem services using two or more case studies. Candidates may evaluate the severity of the different challenges in each case and analyse how these challenges interact, such as how habitat loss can heighten vulnerability to invasive species or how pollution can intensify the effects of climate change. Candidates may evaluate how governance quality, technological capacity, and socio-economic priorities mediate the extent of negative impacts, acknowledging contexts in which human activities have led to the recovery or enhancement of ecosystem services.

4 Evaluate the view that the impacts of TNCs vary spatially. [20]

Candidates could approach the question by making a judgement on whether the impacts of TNCs vary significantly across different spatial contexts. Candidates could evaluate these spatial variations by considering economic, social, and environmental impacts in both host and home countries. They could also analyse how the impacts differ across spatial scales by examining the experiences of different stakeholders in each context, such as by gender, education level, and employment sector. Candidates could also analyse the influence of the power relations between the state and the TNC, which may have contributed to the variations in the impacts.

Candidates could also approach the question by making a judgement on whether positive impacts are greater in some contexts than others through a consideration of two or more case studies. Candidates should consider a range of economic, social and environmental impacts. Candidates could analyse contextual factors such as the development level of the host economy, the stage at which it is connected within the global production network of TNCs, the nature of TNC activities, the host country's governance, and the power relations between the state and the TNC, which may have contributed to the variations in the impacts.

Cluster 2 Tropical Environments

5 ‘Floods in the tropics can be effectively managed.’ Evaluate this statement. [20]

Possible Approaches

Candidates could approach the question by making a judgement on whether floods in the tropics can be effectively managed through a consideration of various flood management strategies (hard engineering measures such as dams, levees and channelisation; soft engineering measures such as afforestation, zoning and floodplain restoration; as well as non-structural strategies such as forecasting, community preparedness and emergency responses). Candidates could evaluate the extent to which these strategies have been successful in reducing the impacts of floods on people, property, and the economy in tropical contexts. Candidates could also evaluate how limitations of flood management strategies—such as high financial costs, maintenance issues, environmental trade-offs, and unequal access to protection—mean that floods in the tropics may not always be effectively managed. Candidates may highlight how ineffective strategies or poorly enforced policies can still result in high levels of flood risk and vulnerability, particularly in developing tropical countries.

Candidates could also approach the question by making a comparative judgement on whether floods can be effectively managed in some tropical contexts more than others, through a consideration of two or more case studies. Candidates could examine contextual factors such as the intensity of rainfall, topography, degree of urbanisation, level of development, governance, and institutional capacity. These factors may contribute to differences in how effective management strategies are in reducing flood risks and impacts in the tropics.

6 ‘Weathering is the most dominant process in the tropics.’ Evaluate this statement. [20]

Possible Approaches

Candidates could approach the question by making a judgement on whether weathering is indeed the most dominant process in the tropics. Candidates could consider the intensity of chemical weathering in the humid tropics and physical weathering in the arid tropics and places of high altitude. Candidates could evaluate how these processes contribute to distinctive landforms in the respective regions in which they operate. Candidates could also evaluate the dominance of other geomorphic processes operating in the tropics e.g. fluvial processes and mass movement through a consideration of criteria such as the spatial and temporal extent of their influence in the humid and arid tropics.

Candidates could also approach the question by making a comparative judgement on the dominance of weathering relative to other processes in different contexts. For example, in humid tropics, chemical weathering may dominate, while in tectonically active or high-relief tropical regions, fluvial erosion or mass movement may play more significant roles. Candidates could also evaluate temporal variation, where weathering dominates long-term landscape development but short-term hazards (e.g. floods, landslides) are governed by other processes.